FINITE PRESENTATION RELATIVE TO A BASE RING CAN BE GLUED ALONG A FINITE COVER BY PRINCIPAL OPENS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes a period before the sentence opening 'such that y_i maps to f_i' at proof line 21146; the punctuation is reproduced unchanged. The five references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Finite presentation of a module relative to a base ring and its behaviour under localization. used at proof lines 21142-21143 and 21154-21157. Is the property being glued and the input that makes the localized modules finitely presented over the localized rings.
- The cover criterion for finite presentation of a module. invoked at proof lines 21157-21158. Turns the local finite presentation of the M_{f_i} into finite presentation of M.
- The auxiliary algebra P = R[x, y, z]/(sum y_i z_i - 1) realizing the cover generators and their unit relation. constructed at proof lines 21145-21153. Provides the ring over which M becomes a finite module and in which the final vanishing argument is run.
- A module with vanishing localizations at elements generating the unit ideal is zero. used at proof lines 21171-21174. Is the final step closing K/K' = 0.

Source reference. Original lemma 00EO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 542–542. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0519: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6527–6527. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 21015: sources/stacks-more-algebra/pinned-original-source.tex, source lines 21015–21015. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 20923: sources/stacks-more-algebra/pinned-original-source.tex, source lines 20923–20923. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $R \to A$ be a finite type ring map. Let M be an A-module. Let $f_1, \ldots, f_r \in A$ generate the unit ideal. The following are equivalent*

1. *each M_{f_i} is finitely presented relative to R, and*
2. *M is finitely presented relative to R.*

Proof. The implication (2) $\Rightarrow$ (1) is in Lemma 21015. Assume (1). Write $1 = \sum f_i g_i$ in A. Choose a surjection $R[x_1, \ldots, x_n, y_1, \ldots, y_r, z_1, \ldots, z_r] \to A.$ such that y_i maps to f_i and z_i maps to g_i. Then we see that there exists a surjection

$$P = R[x_1, \ldots, x_n, y_1, \ldots, y_r, z_1, \ldots, z_r]/(\sum y_i z_i - 1) \longrightarrow A.$$

By Lemma 20923 we see that M_{f_i} is a finitely presented A_{f_i}-module, hence by Algebra, Lemma 00EO we see that M is a finitely presented A-module. Hence M is a finite P-module (with P as above). Choose a surjection $P^{\oplus t} \to M$. We have to show that the kernel K of this map is a finite P-module. Since P_{y_i} surjects onto A_{f_i} we see by Lemma 20923 and Algebra, Lemma 0519 that the localization K_{y_i} is a finitely generated P_{y_i}-module. Choose elements $k_{i,j} \in K$, $i = 1, \ldots, r$, $j = 1, \ldots, s_i$ such that the images of $k_{i,j}$ in K_{y_i} generate. Set $K' \subset K$ equal to the P-module generated by the elements $k_{i,j}$. Then K/K' is a module whose localization at y_i is zero for all i. Since $(y_1, \ldots, y_r) = P$ we see that $K/K' = 0$ as desired. $\square$